UNIVERSITY OF EDINBURGH

COLLEGE OF SCIENCE AND ENGINEERING

SCHOOL OF INFORMATICS

INFR11171 DATA ANALYTICS WITH HIGH PERFORMANCE COMPUTING

Friday 13th May 2022

13:00 to 15:00

Answers must be submitted to Turnitin by 16:00

INSTRUCTIONS TO CANDIDATES

1. **Note that there are THREE COMPULSORY 25-MARK questions. Different sub-questions may have different numbers of total marks. Take note of this in allocating time to sub-questions.**
2. **THIS EXAMINATION IS AN OPEN-BOOK ASSESSMENT. You may refer to material from your notes, course material, or beyond to assist you. You should not copy any text or images into your answer however as your answer must remain your own work. If you refer to material from outside the course it must be referenced properly.**
3. **THIS IS A REMOTE EXAMINATION. As stated in the Own Work Declaration, for the duration of the assessment you must not communicate with any other person about your work either electronically or by word or sign, nor let your work be seen by any other person.**
4. **Please refer to guidance in Learn under Examination Information.**
5. **CALCULATORS MAY BE USED IN THIS EXAMINATION.**

EPCC Courses

Convener: Dr Andreas Pieris
External Examiner: Prof Matt Probert

THIS EXAMINATION WILL BE MARKED ANONYMOUSLY

1. (a) In the three scenarios given below consider whether ETL or ELT would be most appropriate and justify your answer:

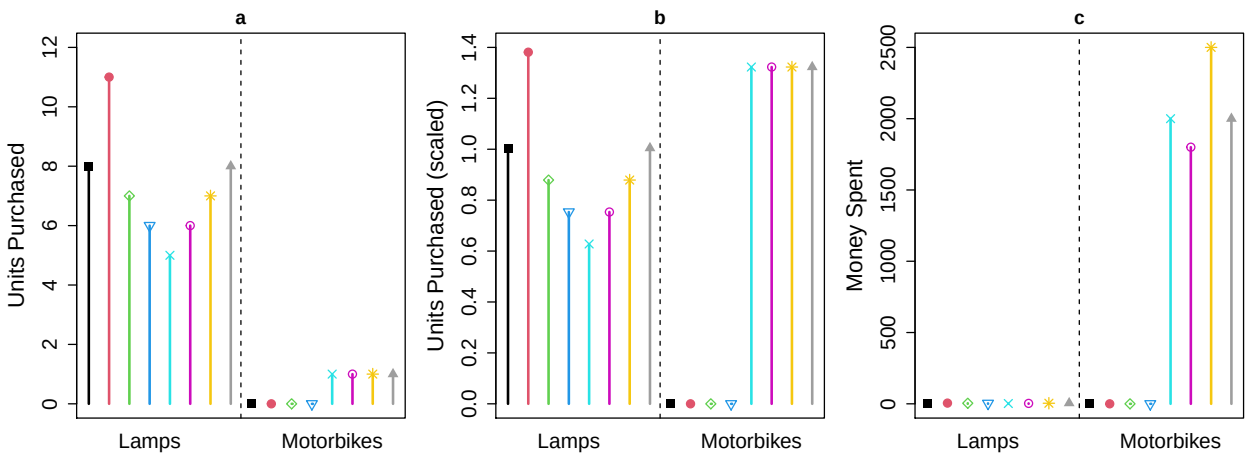
   i. A team within your company wants to detect anomalous insurance claims.

   ii. An organisation wishes to keep a long-term data archive for analysis.

   iii. A secure data source houses highly sensitive data, including individuals' personal information

   [*3 marks*]

   (b) Consider an online merchant only providing two products: lamps and motorbikes. The plots below illustrate the purchases of eight different shoppers where each shopper is depicted by a specific colour and symbol. Plot (a) illustrates the number of lamps and motorbikes purchased by each shopper. Plot (b) illustrates the same information but after scaling each variable by its standard deviation. Lastly, plot (c) shows the amount of money spent by each shopper on lamps and motorbikes.



   Consider you're implementing $k-$means clustering, with $k = 2$, in each of the three cases above. Briefly, explain in general, how the scale/units of measurement of the features could affect k-means clustering. Specifically for the above example, describe in words the results you expect to get in each case and explain why. [*8 marks*]

   (c) A materials science research group requires your help to analyse some data from simulations of a new material. A small sample of their data is shown below and consist of the following three variables: "Type", indicating the category of an observation; "Distance" which is measured in their material and "Shift", a further property of the material.

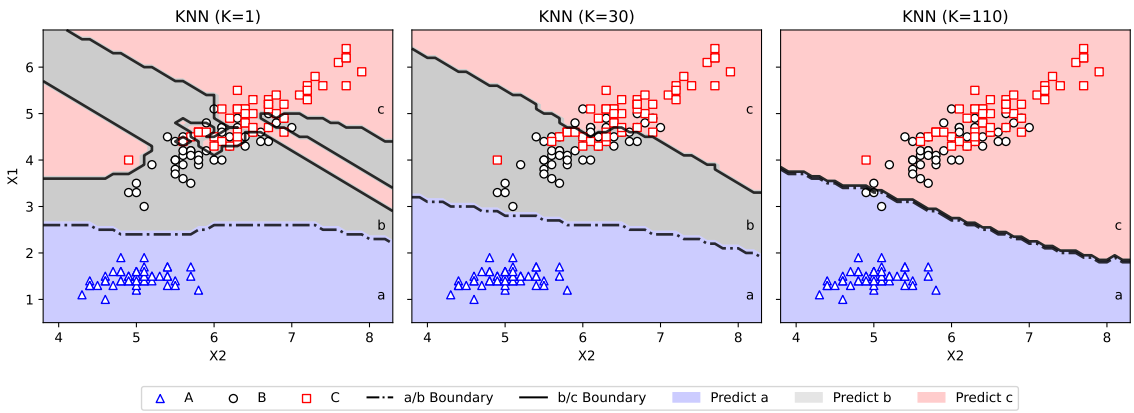
   i. The chemists in the group are interested in finding a way to decide on the "Type" of a new observation using the above data. How can you do that? Mention what approach/method you'd choose, which variable/variables would be used and how. [*2 marks*]

Type	Distance (pm)	Shift (ppm)
1	98.4	2.86
2	104.3	11.62
1	97.8	1.85
3	106.9	12.53
1	98.9	3.46

ii. What plots would you need to look at as part of your data exploration and why? Keep in mind your end goal and how each variable is helping you towards this. [*4 marks*]

iii. The chemists looked into the performance of the model on the training data and found that it wasn't performing so well. However, they realised, that the performance improved after removing some data points. Discuss this approach. Explain how this could affect the performance of the new model. [*3 marks*]

(d) A K-NN classifier has been trained to a dataset with two features, X_1, X_2, and one target variable with three classes. The plots below show scatterplots of the features for the training data. The true class of each data point, A, B and C, is denoted by triangle, circle and square markers respectively. The dashed and straight lines show the classification boundaries of the classifier when trained with $K = 1$, $K = 30$ and $K = 110$. These boundaries indicate how new data points will be classified as indicated by the lower-case labels a, b and c. For example, data points falling within the red area a will be classified as a. Consider that there are 50 examples of each label in the training data.



i. Considering how K-NN works, and with reference to the above plots if helpful, discuss how K affects the classification and the classification

boundaries. In doing so make sure to also discuss the two plausible extremes for the value of K? [*3 marks*]

ii. If $K = 110$ is used to classify new instances of data containing equal numbers of A, B and C instances, it has an overall accuracy of 67%. With reference to the above plot, explain why this is the case and your working. [*2 marks*]

2. DICOM is the international standard data format produced by medical imaging devices. Each DICOM file can contain both the image data and a collection of metadata. The metadata contain information about the patient who was imaged and the properties of the image which was captured.

   (a) When a patient undergoes a CT scan, many image 'slices' are captured. The DICOM files produced by the scanner will contain metadata for the whole scan, and also for each individual image slice.

   Metadata captured from 3 different imaging devices are displayed in the tables below; the first containing scan data and the second containing image data.

Scan ID	**Scan Date**	**Scan Type**	**Image Count**	**Date Requested**
1a	"20220314"	CT	"2"	"2022-2-24"
2a	"20220310"	CT	"1"	"2022-3-12"
3a	"20220412"	MRI	"3"	"2022-4-10"

Image ID	**Scan ID**	**Scan Description**	**Image Date**
1a	1a	Head CT	"20220314"
2a	1a	Head CT	"20220314"
3a	2a	Chest_CT	"2022-03-10"
4a	3a	chest mri	"20220412"
5a	3a	chest mri	"20220412"

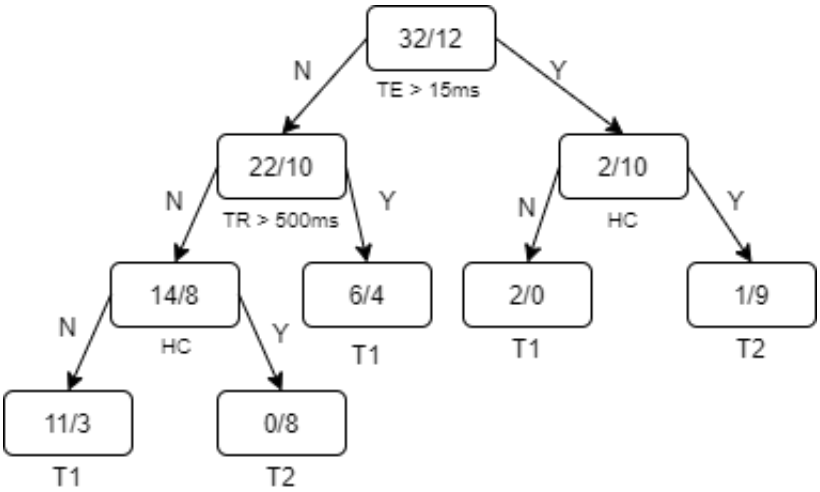
   i. DICOM can be viewed as a *raw* data format. Provide two examples of *raw* data from the tables above that are not *technically correct*. For each, identify a cleaning step which could be applied to that data. [*4 marks*]

   ii. What types of *data consistency* are violated, and how? [*4 marks*]

   (b) *The remainder of this question does not relate to the metadata tables in part a).*

   A researcher wishes to extract and analyse metadata from a dataset of over 1 million raw DICOM images collected from hospitals across Scotland. They are initially interested in analysing the frequency of two types of MRI scan, which they believe can be found in the *Scan Description* metadata fields of each file. Describe the steps you would take to tackle this problem, stating how parallelism can be utilised in each step. [*8 marks*]

   (c) After initial analysis, the researcher has found *Scan Description* to be an unreliable indicator of the scan type. Three new metadata fields have been chosen to build a *Decision Tree* classifier: *Echo Time (TE)*, *Repetition Time (TR)*, and *High Contrast (HC)*. *TE* and *TR* are numeric variables, and *HC* is a boolean variable. The two possible scan types are *T1* and *T2*. Using a labelled subset of the data, the following tree has been built.



i. What scan type will an image with $(TE = 13ms, TR = 550ms, HC = N)$ be classified as having? [*1 mark*]

ii. Taking *T1* to be the "positive" class, create a *confusion matrix* and calculate the *True Positive Rate* and *True Negative Rate* for this classifier. [*3 marks*]

iii. Justify why *HC* has been used twice in the tree, despite being a boolean variable. [*1 mark*]

iv. Describe how the performance of two binary classifiers can be compared. [*4 marks*]

3. (a) In each of the below examples consider what could be an appropriate distribution for modelling the random variable X. Briefly justify your answer highlighting the key reasons.

   i. X: The total number of functioning devices out of boxes of 50 devices

   ii. X: The value, in thousand pounds, of two bedroom properties in Edinburgh.

   iii. X: The daily number of complaints at the call centre of your energy provider. [*3 marks*]

   (b) A big popular theatre in Edinburgh has asked your help to understand how successful are the events they organise. More specifically, they want to know whether an event will be sold-out or not. For this, you're given a csv file with three variables, namely, Y: showing the status of the event (i.e., if it was sold out or not), X_1 : the type of the event (i.e., if it was a theatrical event or a music event) and $X2$: whether it took place during the weekend or not. After processing the data you bring them together in the below format.

	theatrical event	music event
sold-out	100	90
not sold out	50	20

	week-day	weekend
sold out	80	110
not sold out	40	30

   i. They plan to organise a music event on the last Thursday of May. Use a Naive Bayes classifier to answer whether it is going to be sold-out or not. Show all your working using adequate notation. [*8 marks*]

   ii. For the first time, they consider organising a drawing lesson during June's first weekend and they want to know if it's going to be sold out.

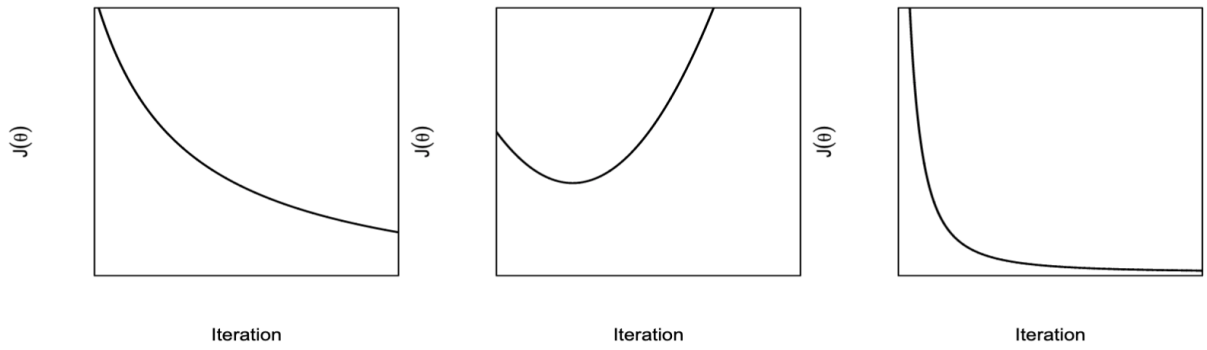
      A. Can you answer this using the provided data and/or the trained classifier? If yes, show and explain how but without making the actual calculation. If not, explain why and suggest how you could approach it. In either case, mention anything that you would like/need to communicate with them based on your approach. [*5 marks*]

   (c) In the process of training various supervised models, we've talked about optimization methods such as gradient descent and stochastic gradient descent.

   i. In this context, what is the learning rate α? [*2 marks*]

   ii. Below you are given three plots of a cost function $J(\boldsymbol{\theta})$ against the number iterations performed by the gradient descent algorithm. Each plot, concerns the same data, training model, same total number number of

gradient descent iterations but different learning rates α_1, α_2, α_3 respectively. Describe and compare the plots in order to describe and explain the effect of the learning rate. Which plot corresponds to the largest and which to the smallest learning rate and why? [*5 marks*]



(d) Which from the below statements are correct?

i. In order to estimate the generalisation error of a model from the training error we need to:

A. First get an estimate of the testing error from the training error

B. Construct a confidence interval for the training error

C. Can't be done

D. Construct a confidence interval for the generalisation error

ii. Suppose we randomly select M testing instances and calculate an error rate R then:

A. I would anticipate the error rate in another set of M instances to also be R.

B. R is the best estimate I have for the error rate for any set of randomly selected instances.

C. R is the best estimate I have for the error rate for any randomly selected set of size M but it doesn't tell me anything for testing sets of other sizes. [*2 marks*]